

Practice problems

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April 12, 2025

Question 1

0.0.1 (a)

Placing n quanta in d boxes

$$g(n) = \binom{n+d-1}{d-1} \quad (1)$$

(b)

The partition function in d dimensions is given by

$$Z_d = Z_1^d \quad (2)$$

$$J = -kT \ln Z_d = -kT \ln Z_1^d = -kTd \ln Z_1 \quad (3)$$

$$\langle N \rangle = - \left(\frac{\partial J_d}{\partial \mu} \right) \quad (4)$$

(c)

$$\ln \Xi = \pm \sum_n \ln \left(1 \pm e^{-\beta(\epsilon_n - \mu)} \right) \quad (5)$$

$$J = -kT \ln \Xi = -kT \sum_n \ln \left(1 \pm e^{-\beta((n+\frac{d}{2})\hbar\omega - \mu)} \right) \quad (6)$$

$$\langle E_1 \rangle = \frac{1}{Z} \sum_{i=1}^{\infty} E_i e^{-\beta E_i} \quad (7)$$

$$Z = \sum_n e^{-\beta(n+\frac{1}{2})\hbar\omega} = \frac{e^{-\frac{\beta\hbar\omega}{2}}}{1 - e^{-\beta\hbar\omega}} \quad (8)$$

$$\langle E_1 \rangle = \frac{1}{Z} \sum_{n=1}^{\infty} \left[\left(n + \frac{1}{2} \right) \hbar\omega \right] e^{-\beta(n+\frac{1}{2})\hbar\omega} = \frac{e^{-\frac{\beta\hbar\omega}{2}}}{Z} \sum_{n=1}^{\infty} n e^{-\beta n \hbar\omega} + \frac{1}{2} e^{-\beta \hbar\omega} \sum_{n=1}^{\infty} n x^n = \frac{-x}{(1-x)^2} \quad (9)$$

$$= \frac{\hbar\omega e^{-\frac{\beta\hbar\omega}{2}}}{Z} \left[\frac{1}{2} \frac{1}{1 - e^{-\beta\hbar\omega}} - \frac{e^{-\beta\hbar\omega}}{(1 - e^{-\beta\hbar\omega})^2} \right] = \frac{\hbar\omega}{2} - \frac{\hbar\omega e^{-\beta\hbar\omega}}{1 - e^{-\beta\hbar\omega}} \quad (10)$$

$$\langle E_d \rangle = d \left(\frac{\hbar\omega}{2} - \frac{\hbar\omega e^{-\beta\hbar\omega}}{1 - e^{-\beta\hbar\omega}} \right) \quad (11)$$

$$C_V = \frac{\partial \langle E_d \rangle}{\partial T} = \frac{\partial \beta}{\partial T} \frac{\partial \beta}{\partial T} = -\frac{1}{kT^2} \quad (12)$$